Mathematical overview: fixed votes, different winners

A majority can go in a circle. With three fixed ranking cards ABC, BCA, CAB (letters read from first to last), A beats B, B beats C and C beats A, each two votes to one. Each card has a consistent ranking, but the three majority results do not. A child can point to the two supporting cards for each result and say, "Each one beats someone and loses to someone."

Waiting can decide a knockout, but need not. Two letters meet first; that winner meets the waiting letter. On these cyclic cards the waiting letter wins. There are only three waiting choices; switching a pair's left and right positions changes no vote. In any three-letter profile with no tied pair result, a letter wins *every* knockout exactly when it beats both others. If it beats both, it wins whether it waits or plays first. If it loses to another letter, meeting that letter first eliminates it. Children can show these possibilities on the mat. With three voters and three letters, only two cases occur: either the votes form a circle and each letter wins exactly when it waits, or one letter beats both others and wins every knockout. No other letter can then win any knockout.

What B needs on the cards. B wins every knockout when first on two or three cards. When first on exactly one card, B wins only if second on both other cards, once below A and once below C. It never wins when first on no card. Each rival needs two cards with B above it: four supporting places in all. A B-first card supplies two places, a B-second card one, a B-last card none. With one B-first card, each other card must supply one different missing comparison; with none, three cards cannot supply four places.

One swap ends this circle. Swapping two letters on one of ABC, BCA, CAB makes the letter that moved up the only knockout winner (all nine swaps checked). Moving up never costs that letter a vote. On these cards every upward swap passes the letter that beat it, turning a 1–2 loss into a 2–1 win; it already beat the remaining letter. This is specific to these cyclic cards, not arbitrary profiles.

Assumptions and limits. Each of three imaginary voters ranks A, B and C once, with no equal ranks. Each votes for the higher of the *two offered* letters, even when its first choice is absent. All three vote, so a pair cannot tie. The cards stay unchanged throughout a knockout. The child's own favorite does not vote. These finite facts do not establish a fairness claim or Arrow's theorem; changing cards between rounds is a different investigation.

What children should discover. Problem 1's pair votes go round in a circle: no letter beats both others. On those cards, Problem 2's waiting letter decides the winner. Problems 3–4 show instead a letter that wins wherever it plays; in Problem 4 it is first on only one card. Problem 5 breaks the circle with one swap. Problems 6–8 ask what forces B to win: two cards with B above A and two with B above C. Grades 2–5 use the shared packet, with upper readiness adding profile design and explanations. The line/median destination is a separate visit (base guide overview, page 1), outside this K–1 route.

Evidence children can give. One completed vote or knockout is an experiment. "The waiting letter always wins" is initially a conjecture about the current cards, not all cards. Trying all three waiting letters settles that finite question; pointing to the cards that force both pair wins explains why a fixed winner survives every agenda. A few unsuccessful builds do not prove impossibility.

Preparation, shared launch and a first visit

Readiness and status. Children recognize A–F, match V1–V3 to tokens, compare higher/lower and count to three; independent reading/writing is unnecessary. An anchored adult reads prompts and mat labels. The partner checks each token as placed; the adult checks the count and asks for a replay when it differs from the key. Children choose ballots, waiting letters and builds. Stagger two pairs' first knockouts. Unpiloted; cutting, print scale, slip fit, two-mat handling, resets and timings are physically unrehearsed.

Print per pair. One six-page `k-1.pdf` (W87-K-v1; partners take turns circling), plus one copy each of `week-87-materials.pdf` (W87-M-v1) pages 1 and 4. Add page 2 for Problems 5–8. Print US Letter, single-sided, Actual Size (100%); check the materials' 100 mm scale. Two K–1 pairs add two copies of each required materials page, beyond the older tables' kits. Materials pages 3 and 5 are not needed.

One pair kit. Materials page 1 supplies three fixed 45 by 80 mm cards, six 28 mm square option tiles (two each A/B/C), and the three 18 mm vote discs V1–V3. Set V4 and V5 aside. Page 4 stays whole: a 170 by 90 mm tally mat with three 54 mm square wells, and a 175 by 100 mm knockout mat. Card votes never use Tie. Page 2 supplies three blank 45 by 80 mm cards and eighteen 21 mm letter slips (six of each letter); only nine slips are needed on the three cards at once. Keep the rest as spares. The roughly 23 mm card rows are tight fits; check real slips before use.

Extra preparation. Allow an additional 15–25 minutes to print, cut and bag two core K–1 kits; add 10–15 minutes for two blank-card/slip sets and 5–10 minutes to rehearse a whole knockout. These are estimates, not measured times. Supply a pencil and scrap paper. Optional matching color dots go on each card/token and the V labels on student pages 1, 3, 4 and 5. Dots and the cover slips described under Problem 1 are untried. For the table demonstration make six temporary 28 mm square labels D,D,E,E,F,F from scrap (about two minutes). No week file prints those movable letters; its D/E/F rankings are pictures, not cut cards.

After the whole-group launch. Let partners handle the real pieces. Revisit the non-task DFE, EFD, FED pictures on student page 1. Label the two real tally wells D and E. Ask one child to place V1's token and a partner to check the higher offered letter; children place the remaining tokens. V3 still votes although its first choice F is absent. Point to the already circled E to show how a winner is recorded. Use the printed input, tally and circled output as the visual reminder. **Before Problem 2, not before Problem 1.** Use the same pictures and temporary labels: First pair D/E, Waiting F; E wins, then F beats E, both 2–1. Match student page 2's visual and read the mat labels. Use one copy of each letter on the knockout mat and the second copy to label tally wells. The first loser goes to Out and the winner to First winner. Then move it and the waiting tile into Final pair. The final loser goes to Out and the winner to Winner. Clear *all three tokens* before the second vote. Reset tiles between knockouts, never the rankings. Remove the temporary D/E/F labels before A/B/C work. This example has one fixed winner, not a cycle, and reveals no task answer.

One-hour route. Allow 5 minutes for movement, 5 for the shared launch, 5 for table handling/recap, 10 for Problem 1, 15 for the knockout recap and Problem 2, 15 for Problem 3, and 5 to share one surprise. Stop earlier if resets still require adult decisions. Problem 4 is a ready pair's continuation; save Problems 5–8 for another visit. Use the small blank frames on pages 5–6 for adult pencil records, not letter slips. The children build on full-size blank cards. Finishing every page is not the goal.

Solutions and held hints: the first route

Adult shorthand such as ABC means first A, second B, third C. Below, page numbers refer to the six-page student packet. Keep solutions out of children's sight.

Problem 1, page 1: three pair votes

Circle **A, B, C**, respectively. A beats B with V1 and V3; B beats C with V1 and V2; C beats A with V2 and V3. Each result is 2–1. Children can show the two supporting cards rather than explain in sentences. After all three results, ask, "Is there a letter that beat both others?" There is none: each has a loss.

Held hints. "Which two letters are on offer?" "Which is higher on this voter's card?" If a child votes for an unavailable first choice, temporarily cover that letter with a small scrap and ask again. Do not discard the voter or turn the ranking upside down.

Problem 2, page 2: the three printed knockouts

Keep the fixed cards ABC, BCA, CAB, not the D/E/F example cards.

First pair	Waiting	First winner	Final vote	Circle
B, C	A	B	A beats B	A
C, A	B	C	B beats C	B
A, B	C	A	C beats A	C

Every vote is 2–1. These are all three waiting choices. Let children notice the pattern; do not announce "waiting wins" first. In each case the first winner loses to the letter that waited. The D/E/F demonstration may also suggest "waiting wins, but F won there because it beats both; Problem 3 separates these reasons.

Held hints. "Who has not played yet?" "Which letters reach the final pair?" If cards or votes change between rounds, check the cut cards against the page 1 picture and clear only the vote tokens. A natural continuation is to reverse the left/right order of one first pair and see whether any ballot changes; it does not.

Problem 3, page 3: CBA, BAC, CBA

Circle **C only**. C beats A and B, each 2–1, using V1 and V3. B beats A 3–0. With A waiting, C beats B, then C beats A. With B waiting, C beats A, then C beats B. With C waiting, B beats A, then C beats B. Thus C wins all three; A and B never win.

Put away the cut fixed cards and vote directly from the full-size pictures on this page; children need not copy them. The two CBA cards are *two voters*, so both tokens count. On scrap, the adult may record one short row per tried waiting letter and its final winner, leaving the agenda choices to children. This prevents repeating an attempt without noticing.

Held hints. "Who is voting on this page?" "Can either other letter beat C?" A child can point to the two cards with C first: neither rival can remove C, whether C waits or plays first. This breaks a universal "waiting wins" conjecture without changing the voting rule.

Problem 4, page 4: ABC, BAC, CAB

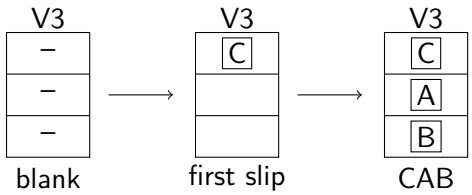
Circle **A only**. A beats B (V1/V3) and C (V1/V2), each 2–1; B beats C (V1/V2), also 2–1. Waiting A gives B then A; waiting B gives A then A; waiting C gives A then A. Only A wins. Each letter is first on one card, so counting first choices alone would miss the result. Again vote from the printed cards, not the old cut set.

Held hints. "If A plays first, who can knock it out?" "If A waits, who could beat it at the end?" After testing, ask how these cards differ from Problem 1: only V2 has exchanged A and C. That comparison prepares later card changing without solving Problem 5's different target.

Return visit: change and build the voters

Restart and route. For 3–5 minutes, handle the cut V1–V3 cards and replay one Problem 1 vote. This is the K–1 launch if older tables are now on the separate line visit. If Problem 4 was skipped, allow about 10 minutes for it; compare its V2 with Problem 1 to show one A/C swap and a new winner A, without solving Problem 5's target C.

Before building. Beside the cut V3 card, put a C slip in the blank V3's first row, then A and B below to make the same CAB. It casts the same votes as the printed card. A card cannot contain two Bs: each letter appears once. Let children handle the slips, then clear the card. Compact adult reminder (not a cutting template):



Core and continuations. Allow about 15 minutes for Problem 5 and 10 for Problem 6. Offer Problem 7 only after a quick Problem 6 success, holding back its hints; Problem 8 is an adult-held loop conversation for ready pairs. Leave time to share a build. These are flexible estimates; finishing is not the goal.

Problem 5, page 5: one swap to make C win

Take the cut V1–V3 cards out again: ABC, BCA, CAB. Build the replacement on the blank card with the same V label. Exactly three swaps work: **V1 ABC to ACB** (swap B/C); **V1 ABC to CBA** (swap A/C); or **V2 BCA to CBA** (swap B/C). C beats B 2–1 in all three; C beats A 2–1, 3–0 and 2–1, respectively. Thus C wins every knockout. All other cards stay fixed. Make only the replacement card with slips and put the original aside. Reset to the original three cards before testing another swap.

The working swaps are exactly those that move C up: two on V1 and one on V2; V3 already has C first. The overview explains why all nine single swaps make their upward-moving letter the only winner. Children need one working construction, not this adult catalog. **Held hints:** "Which rival stops C on the original cards?" "Which card could change its vote between B and C?" C already beats A; the changed ranking must also let C beat B without losing that advantage.

Problem 6, page 5: build a fixed winner B

One example is **BAC, BCA, ABC**: B beats A 2–1 and C 3–0, and wins all three knockouts. Repeated cards are allowed; the older shared packet's "three different cards" rule does not apply here. Any legal set with B first on two or three cards works, since those cards support B against either rival. Accept a different build after both pair votes (B against A and B against C) go to B; pointing to the supporting cards is the explanation; putting B first everywhere is a valid entry, not a failure of imagination.

Held hints. "Which two cards could keep voting for B against either other letter?" "Does your set work when B plays first as well as when B waits?" Let the child arrange the slips; the adult records the final ranking in the small frames. To deepen a quick success, move to the next problem rather than demanding many similar builds.

Later solutions, observations and integration notes

Problem 7, page 6: B first exactly once

It can be done. The only working sets, ignoring which voter receives which card, are

$$\{ABC,\ BAC,\ CBA\} \qquad \text{and} \qquad \{ABC,\ BCA,\ CBA\}.$$

In either, B beats A and C, both 2–1. Assign the three cards to V1–V3 in any order. The B-first card supplies one vote against each rival. To get a second vote in both comparisons, the other two cards must put B second, one below A (ABC) and one below C (CBA). A B-last card supplies neither vote. The B-first card may be BAC or BCA; these exhaust the possibilities.

Held hints. "Which of the two rivals still beats B?" "What can a card with B second help B beat?" Do not hand over the answer after a few random failures. Children can keep their one B-first card and rearrange the others. A ready pair can find both working sets; this is optional, not a new written task.

Problem 8, page 6: B never first

Impossible. Draw two separate loops on scrap, labelled with spare A and C tiles. Explain that the A loop is for cards voting B over A, the C loop for B over C. Sort each voter's token by its card: B second fits one loop; B last fits neither. With B never first, no token qualifies for both. For B to beat both rivals, each loop would need two different voter tokens, but only three exist. This is a grouping proof, not two simultaneous votes. Children may simply point to the shortage; no written counting argument is required.

Held hints. "Can a card with B not first put B above both A and C?" "Can both loops have two of our three tokens without using one twice?" Put the tokens back by the tally mat afterwards. Failed builds alone are not the explanation; the shortage applies to every legal build.

What to observe. Record the problem each pair reached, whether partners kept "the higher of the two on offer" when a top choice was absent, slip fit and two-mat handling, and any vote, waiting letter or build selected by the adult. See the base guide, page 6, for the general observation list.

Where it goes. Once the pages are approved, this section replaces "K–1 route (adult-led)" on page 1 of `week-87-facilitator.pdf` (W87-F-v2). Update page 2's "Prepare three older-table pair kits" to add two K–1 kits; page 6's Print map to include this six-page packet and section; and the Week 87 catalog record. These are later integration instructions: existing files remain untouched. The older-band overview and solutions stay in place.